

ASSIGNMENT #6

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: _____ **Diana Awimrin** _____
Due Date: Saturday April 11 at 12 PM (Noon)

Number of points.....100
Your score.....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 6A: Design of Feedback Controller for Single Loop Disturbance Rejection

In order to obtain data for this problem, the Athena Code from Assignment 3 is slightly modified to fit the criteria of this assignment. First, the residence time of the cooling jacket water is modified to be 25% of the residence time of the reactor and the values of UA and C_p change to $60737 \frac{kJ}{hr K}$ and $116.76 \frac{kJ}{kmol K}$, respectively. To account for the process gain of one and a time lag of 5 minutes the transfer function and its Laplace inverse seen in *Equation 1* is used. T_m is the measured temperature and T_R is the reactor temperature.

$$\frac{T_m}{T_R} = \frac{1}{5s + 1} \rightarrow 5 \frac{dT_m}{dt} + T_m = T_R \quad (1)$$

Following the equation above, in the Athena code an additional initial condition, $U(7)$, is added and is equated to a random value for now. Under the model equations T_m is equated to $U(7)$ and $F(7)$ is equated to $T_R - T_m$. Under the coefficient matrix, $E(7)$ is equated to the coefficient of 5. These modifications can be seen in the Athena Code in **Appendix A**. The code is then executed, and the steady state values are copied and pasted into the original initial conditions. To ensure steady state, the initial condition of the measured temperature, $U(7)$, needs to be equivalent to the initial condition of the reactor temperature, $U(5)$. Once the new values are in, the code is executed once again, and a graph is plotted with a steady state temperature of 346.44K and is shown in *Figure 1*.

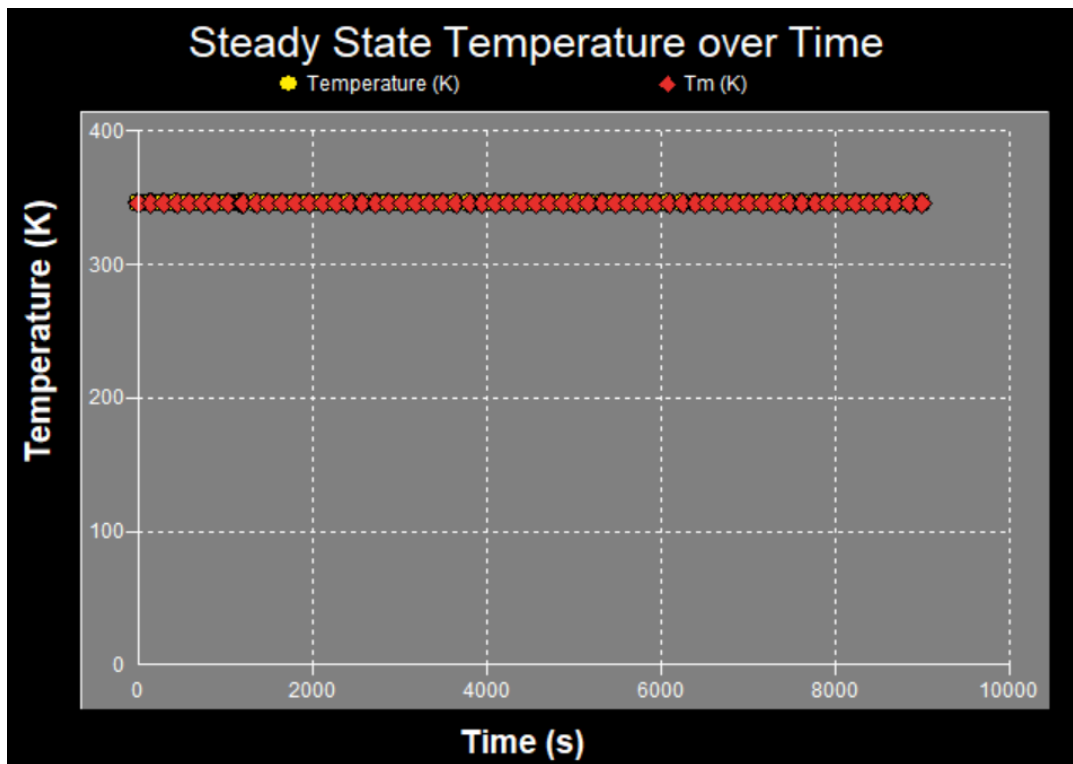


Figure 1: Steady State Temperature over Time

With the process now running at steady state, we want to see how the system will respond to the manipulated variable. This process changes by manipulating the flowrate of the cooling water. In order to perform this step test, an if-statement is incorporated into the code with the cooling water flow rate starting at $0.367 \frac{m^3}{min}$ and ending at $0.4 \frac{m^3}{min}$ to prevent a maximum temperature change of $\pm 5^\circ C$. These changes are seen in the Athena code in **Appendix A**. The code is then executed and a graph is plotted showing the reactor temperature compared to the measured temperature with a slight time delay as shown in *Figure 2*. The temperatures start off at 346.44K and drops off to 345.45K.

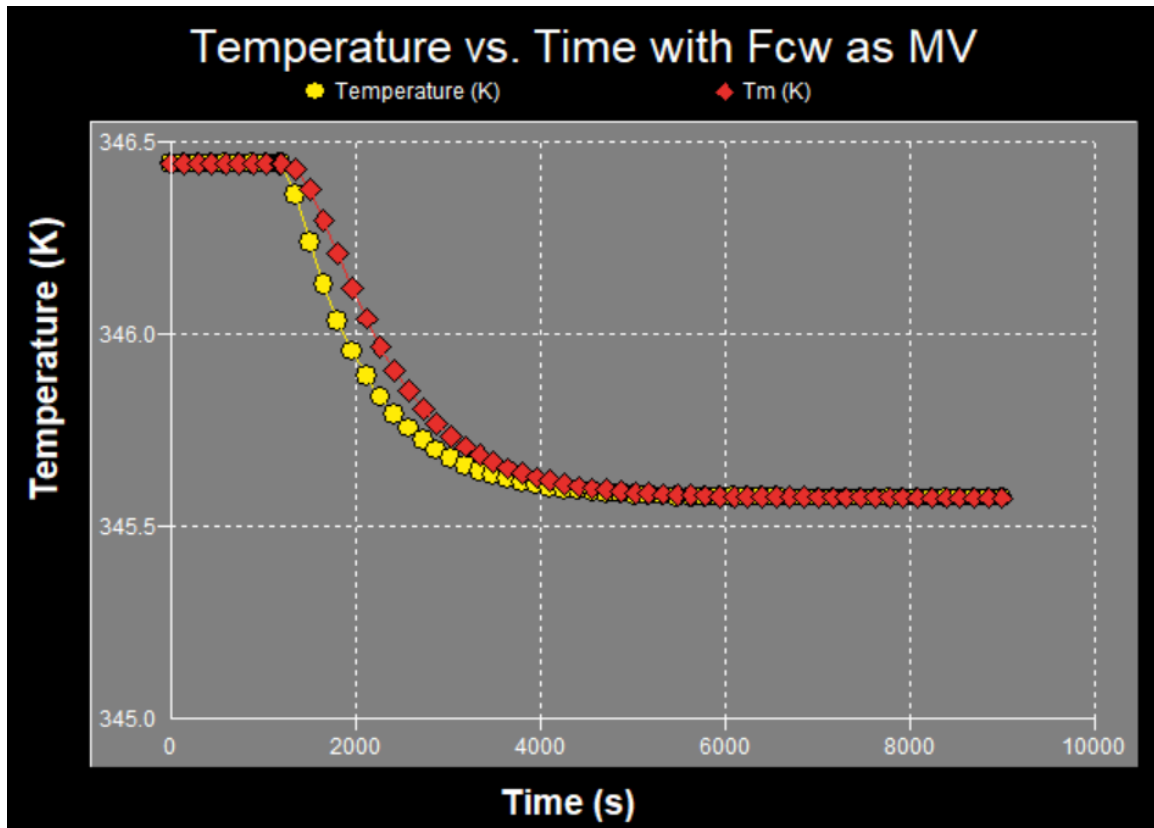


Figure 2: Temperature vs. Time with Fcw as MV

Now, the first disturbance will be incorporated into the code which will be done by manipulating the temperature of the cooling water. In order to perform this step test, an if-statement is incorporated into the code with the temperature of cooling water starting at 298.15K and ending at 302.15K rate to prevent a maximum temperature change of $\pm 5^\circ\text{C}$. These changes are seen in the Athena code in **Appendix A**. The code is then executed and a graph is plotted showing the reactor temperature compared to the measured temperature with a slight time delay as shown in *Figure 3*. The temperatures start off at 346.44K and drops off to 348K.

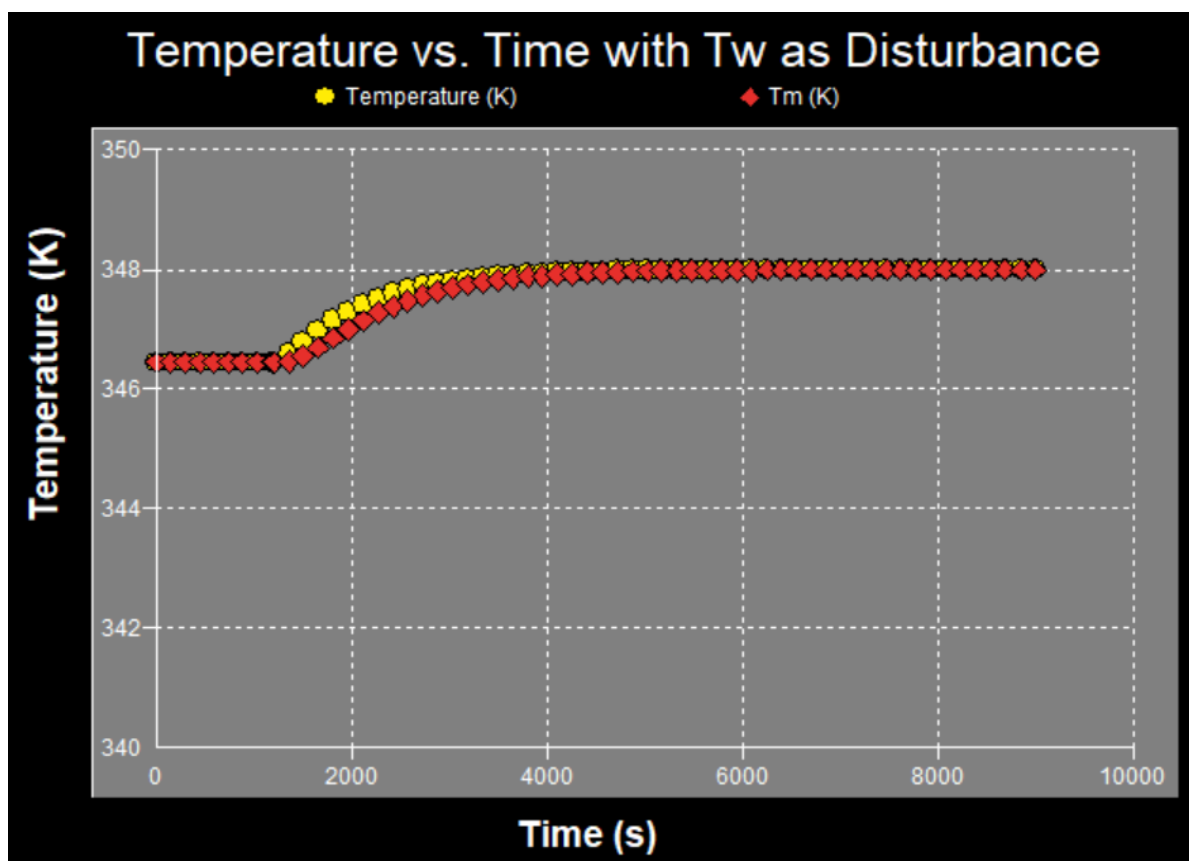


Figure 3: Temperature vs. Time with T_{water} as Disturbance

Next, the second disturbance will be incorporated into the code which will be done by manipulating the temperature of the feed. In order to perform this step test, an if-statement is incorporated into the code with the temperature of feed starting at 298.15K and ending at 303.15K rate to prevent a maximum temperature change of $\pm 5^\circ\text{C}$. These changes are seen in the Athena code in **Appendix A**. The code is then executed and a graph is plotted showing the reactor temperature compared to the measured temperature with a slight time delay as shown in *Figure 4*. The temperatures start off at 346.44K and drops off to 348K.

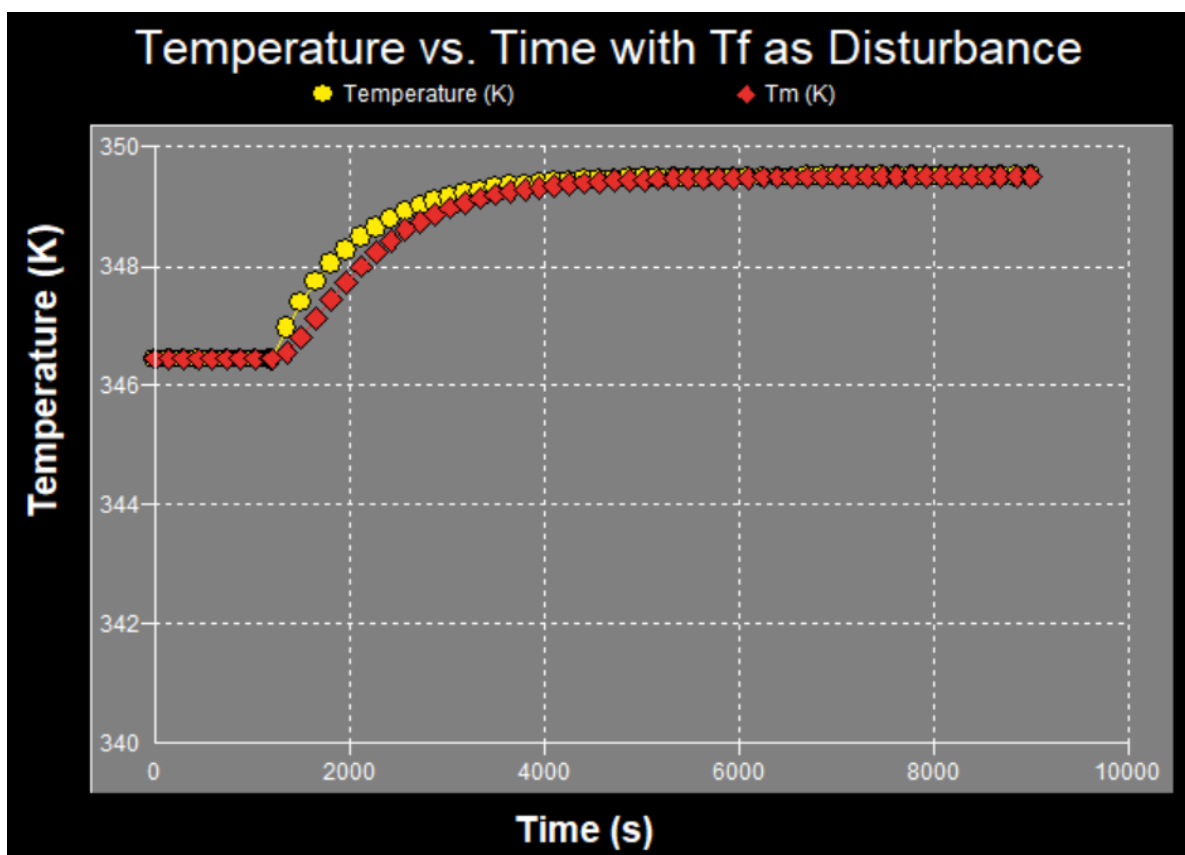


Figure 4: Temperature vs. Time with T_{feed} as Disturbance

Now, an Excel table can be created to have columns made up of the time and the newly obtained data from each of the three step tests performed previously. The FOTD is then calculated for each set of data using *Equation 2* with the step size difference changing for each set of step test data. When the flow rate of the cooling water was incorporated the step size difference was (0.4 - 0.367). When the cooling water temperature was incorporated the step size difference was (302.15 - 298.15)K. Lastly, when the feed temperature was incorporated the step size difference was (303.15 - 298.15)K.

$$Y(t) = T_r + K_p * (\text{Step Size Diff}) * \left(1 - e^{-\frac{t-\theta}{\tau}}\right) \quad (2)$$

The RSQ is then calculated for by taking the difference squared between the new data and the FOTD for each of the respective step tests performed and then summed. Excel's Solver is then used in order to solve for the K_p , τ , and θ values. In Solver, the RSQ is minimized by changing the K_p , τ , and θ values. This method is done for all three step tests. The solver answers are shown below for the step tests of the cooling water flow rate, the cooling water temperature, and the reactor temperature in *Table 1, 2 and 3*, respectively.

Table 1: Values obtained when Cooling Water Flow Rate Changed

FOTD (F _{cw})		
Variable	Solver Answers	Original Guess
K _p (m ³ /min)	-26.48703795	-18
Tau (min)	16.15997593	15
Theta (min)	4.624239201	5
RSQ	0.0049	

Table 2: Values obtained when Cooling Water Temperature Changed

FOTD (T _{cw})		
Variable	Solver Answers	Original Guess
K _p (m ³ /min)	0.389189502	1
Tau (min)	14.74728557	10
Theta (min)	6.122663001	5
RSQ	0.0259	

Table 3: Values obtained when Feed Temperature Changed

FOTD (T _f)		
Variable	Solver Answers	Original Guess
K _p (m ³ /min)	0.768189085	1
Tau (min)	15.84067309	10
Theta (min)	3.60874229	5
RSQ	0.0416	

Now, with the process parameters known (shown in Table 1 above) the K_c , τ_I , and τ_D values are solved for both PI and PID controllers each of the following methods: Ziegler-Nichols, Cohen-Coon, Lambda tuning, and Ultimate Cycling tuning. The equations used are shown below (Equation 3 - Equation 25). These will all be interactive controllers. The values calculated from these equations are shown in Table 4 – Table 12 in **Appendix B**.

Ziegler-Nichols		
PI	K_c	$\frac{0.9}{SM K_p} \left(\frac{\tau}{t_d} \right) \quad (3)$
	τ_I	$3.33 t_d \quad (4)$
PID	K_c	$\frac{1.2}{SM K_p} \left(\frac{\tau}{t_d} \right) \quad (5)$
	τ_I	$2 t_d \quad (6)$
	τ_D	$0.5 t_d \quad (7)$

Cohen-Coon (Non-Interactive)		
PI	K_c	$\frac{0.9}{SM K_p} \left(\frac{\tau}{t_d} + 0.092 \right) \quad (8)$
	τ_I	$3.33 t_d \left(\frac{\tau + 0.092 t_d}{\tau + 2.22 t_d} \right) \quad (9)$
PID	K_c^{NI}	$\frac{1.35}{SM K_p} \left(\frac{\tau}{t_d} + 0.185 \right) \quad (10)$
	τ_I^{NI}	$2.5 t_d \left(\frac{\tau + 0.185 t_d}{\tau + 0.611 t_d} \right) \quad (11)$
	τ_D^{NI}	$0.37 t_d \left(\frac{\tau}{\tau + 0.185 t_d} \right) \quad (12)$

For the Cohen-Coon method, the equations above are used to find K_c , τ_I , and τ_D for the non-interactive controller. In order to solve for the interactive values, *Equations 13, 14 and 15* are used. The values needed for these three equations are obtained from *Equations 10, 11 and 12*.

Cohen-Coon (Interactive)		
PID	K_c	$K_c^{NI} \left(\frac{1}{2} + \sqrt{\frac{1}{4} - \left(\frac{\tau_D}{\tau_I} \right)^{NI}} \right)$ (13)
	τ_I	$\tau_I^{NI} \left(\frac{1}{2} + \sqrt{\frac{1}{4} - \left(\frac{\tau_D}{\tau_I} \right)^{NI}} \right)$ (14)
	τ_D	$\frac{\tau_D^{NI}}{\left(\frac{1}{2} + \sqrt{\frac{1}{4} - \left(\frac{\tau_D}{\tau_I} \right)^{NI}} \right)}$ (15)

Lambda Tuning		
PI	K_c	$\frac{\tau}{K_p(\tau_c + \theta)}$ (16)
	τ_I	τ (17)
PID	K_c	$\frac{\tau}{K_p(\tau_c + \theta)}$ (18)
	τ_I	τ (19)
	τ_D	$\frac{\tau}{2}$ (20)

Ultimate Cycling		
PI	K_c	$0.45 \left(\frac{K_u}{2} \right)$ (21)
	τ_I	$0.83 P_u$ (22)
PID	K_c	$0.6 \left(\frac{K_u}{SM} \right)$ (23)
	τ_I	$0.50 P_u$ (24)
	τ_D	$0.125 P_u$ (25)

For the Ultimate Cycling method, K_u and P_u are calculated by plugging in the process parameters into the ChE341 Calculator in the PID Controller tab. Next, the gain of controller 1 is manipulated until the graph shows oscillations of about the same amplitude. Omega (ω) is then calculated with *Equation 26*, with P_1 and P_2 being the period between one wavelength. The omega value, -0.382, is then plugged into the ultimate gain tab and K_u and P_u are calculated to be -0.232 and 16.739, respectively and shown in *Table 12* in **Appendix B**.

$$\omega = \frac{2\pi}{P_1 - P_2} P_u \quad (26)$$

Next, each method's PI and PID values shown above are plugged into the ChE341 Calculator in the controller sections. Along with those values, the K_p , τ , and θ values from *Table 2* and *Table 3* are plugged in the process variable section to account for when the cooling water temperature was the disturbance versus when the feed temperature was the disturbance. The graphs made from plugging in these values are shown in *Figure 5 – Figure 12* in **Appendix C**.

After analyzing these graphs, it can be concluded that the **Lambda Tuning Method** is the best method to use for this process. This is because these graphs stay within in the set point (no dip into the negatives) and their curves are much smoother than the rest of the controllers. This means that the PI and PID controllers using the Lambda Tuning Method will be able to perform their job at rejecting disturbances and stabilizing the temperature of the process in the most efficient manner.

Appendix A: Athena Code

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

Global Qfeed, Qcool, FAA, FEtOH, FEAC, FWAT, Tfeed, Twater, Tcool, Tc As Real

Qfeed = 7.5758 !L/s

Qcool = 431726 !J/s

FAA = 66.75 !mol/s

FEtOH = 64.592 !mol/s

FEAC = 0 !mol/s

FWAT = 3.506 !mol/s

Tfeed = 298.15 !K

Twater= 298.15 !K

Global Vol, UA As Real

Vol = 10000 !L

UA = 16871.39 !J/s K

Global rho, Cp, DHR1 As Real

rho = 17.8 !mol/L

Cp = 116.76 !J/mol K

DHR1 = -29070 !J/mol

Global kro, kfo, Ero, Efo As Real

kfo = 1.9E+08 !mol/L s

kro = 5.0E+07 !mol/L s

Ero = 59500 !J/mol

Efo = 59500 !J/mol

Global rhowater, cpwater, MWwater, CvX, Fcw, VolJ As Real

rhowater = 997 !g/L

cpwater = 75 !J/mol K

$$MW_{\text{water}} = 18.02 \quad !\text{g/mol}$$

$$CvX = 35$$

$$F_{cw} = CvX * \sqrt{\Delta P} * (3.79/60) \quad !\text{L/s}$$

$$VolJ = F_{cw} * 1319.99 * (0.25 * 0.5) \quad !\text{L}$$

@Initial Conditions

!Steady State

$$U(1) = 3.16299E+00 \quad !\text{mol/L}$$

$$U(2) = 2.87813E+00 \quad !\text{mol/L}$$

$$U(3) = 5.64796E+00 \quad !\text{mol/L}$$

$$U(4) = 6.11075E+00 \quad !\text{mol/L}$$

$$U(5) = 3.46444E+02 \quad !\text{K}$$

$$U(6) = 3.17789E+02 \quad !\text{K}$$

$$U(7) = 3.46444E+02 \quad !\text{K}$$

@Model Equations

Dim kf, kr, Rate, CAA, CEtOH, CEAC, CWat, Tr, Tm As Real

$$CAA = U(1)$$

$$CEtOH = U(2)$$

$$CEAC = U(3)$$

$$CWat = U(4)$$

$$Tr = U(5)$$

$$Tc = U(6)$$

$$Tm = U(7)$$

$$kf = kfo * \exp(-Efo / (Rg * Tr))$$

$$kr = kro * \exp(-Ero / (Rg * Tr))$$

$$\text{Rate} = (kf * CAA * CEtOH) - (kr * CEAC * CWat)$$

!FeedBack Controller

!If (iRange == 1) Fcw = 6.11	!L/s
!If (iRange == 1) Fcw = 6.67	!L/s Uncommented when Fcw step test is done
!If (iRange == 3) Fcw = 2.33	!L/s
!If (iRange == 4) Fcw = 6.12	!L/s
!If (iRange == 1) Twater = 298.15	!K
!If (iRange == 1) Twater = 302.15	!K Uncommented when Twater step test is done
!If (iRange == 3) Twater = 294.15	!K
!If (iRange == 4) Twater = 298.15	!K
!If (iRange == 1) Tfeed = 298.15	!K
If (iRange == 1) Tfeed = 303.15	!K Uncommented when Tfeed step test is done
!If (iRange == 3) Tfeed = 288.15	!K
!If (iRange == 4) Tfeed = 298.15	!K

$$F(1) = F_{AA} - (Q_{feed} \cdot C_{AA}) - (Rate \cdot Vol)$$

$$F(2) = F_{EtOH} - (Q_{feed} \cdot C_{EtOH}) - (Rate \cdot Vol)$$

$$F(3) = F_{EAC} - (Q_{feed} \cdot C_{EAC}) + (Rate \cdot Vol)$$

$$F(4) = F_{WAT} - (Q_{feed} \cdot C_{Wat}) + (Rate \cdot Vol)$$

$$F(5) = ((Q_{feed} \cdot C_p \cdot \rho) \cdot (T_{feed} - T_r)) - (UA \cdot (T_r - T_c)) + ((-DHR1) \cdot Rate \cdot Vol)$$

$$F(6) = (F_{cw} \cdot (\rho_{water}/MW_{water}) \cdot c_{pwater} \cdot (T_{water} - T_c)) + (UA \cdot (T_r - T_c))$$

$$F(7) = -T_m + T_r$$

@Coefficient Matrix

$$E(1) = Vol$$

$$E(2) = Vol$$

$$E(3) = Vol$$

$$E(4) = Vol$$

$$E(5) = \rho \cdot C_p \cdot Vol$$

$$E(6) = (\rho_{water}/MW_{water}) \cdot c_{pwater} \cdot VolJ$$

$$E(7) = 300 \quad !seconds$$

@Solver Options

$$Neq = 7 \quad !Equations$$

$$Npts = 60 \quad !Graph Points$$

$$Tend = 9000 \quad !Time, seconds$$

$$Tstop = 1200$$

$$Param(28) = 1$$

$$VariableNames = Time(s);CAA;CEtOH;CEAC;CWat;Temperature(k);Tc(k); Tm (k)$$

Appendix B: K_c , τ_I , and τ_D values PI and PID controllers for all Methods

Table 4: Z-N Method PI

Ziegler-Nichlos PI	
Kp	-26.487
Tau	16.15998
Td (theta)	4.624239
SM	2
Kc	-0.05937
Ti	15.39872

Table 5: Z-N Method PID

Ziegler-Nichlos PID	
Kp	-26.487
Tau	16.15998
Td (theta)	4.624239
SM	2
Kc	-0.07916
Ti	9.248478
Td	2.31212

Table 6: C-C Method PI

Cohen-Coon PI	
Kp	-26.487
Tau	16.15998
Td (theta)	4.624239
SM	2
Kc	-0.06093
Ti	9.664574

Table 7: C-C Method PID

Cohen-Coon PID	
Kp	-26.487
Tau	16.15998
Td (theta)	4.624239
SM	2
Kc (NI)	-0.09377
Ti (NI)	10.36107
Td (NI)	1.624946
Kc (I)	-0.07551
Ti (I)	8.34309
Td (I)	2.017979

Table 8: Lambda Tuning Method PI

Lambda Tuning PI	
Kp	-26.48704
Tau	16.159976
Theta	4.6242392
SM	2
Kc	-0.037754
TI	16.16

Table 9: Lambda Tuning Method PID

Lambda Tuning PID	
Kp	-26.48704
Tau	16.159976
Theta	4.6242392
SM	2.00
Kc	-0.029354
TI	16.16
TD	2.3121196

Table 10: Ultimate Cycling Method PI

Ultimate Cycling PI	
Kp	-26.487038
Tau	16.159976
Theta	4.6242392
SM	2
Kc	-0.0522
TI	13.89337

Table 11: Ultimate Cycling Method PID

Ultimate Cycling PID	
Kp	-26.487038
Tau	16.159976
Theta	4.6242392
SM	2
Kc	-0.0696
TI	8.3695
TD	2.092375

Table 12: Gain and Period used for UC method

Ultimate Gain and Period	
Ku	-0.232
Pu	16.739

Appendix C: Graphs plotted for each Controller Method

Note: The blue graph lines are the PI controllers and the red graph lines are the PID controllers

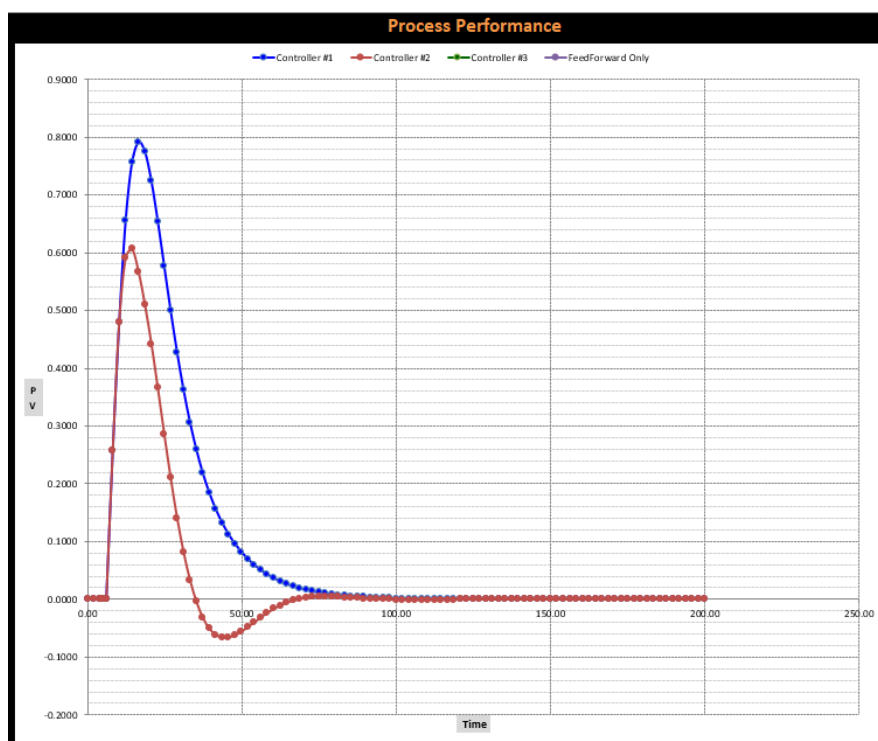


Figure 5: Ziegler-Nichols Method for Cooling Water Temperature Disturbance

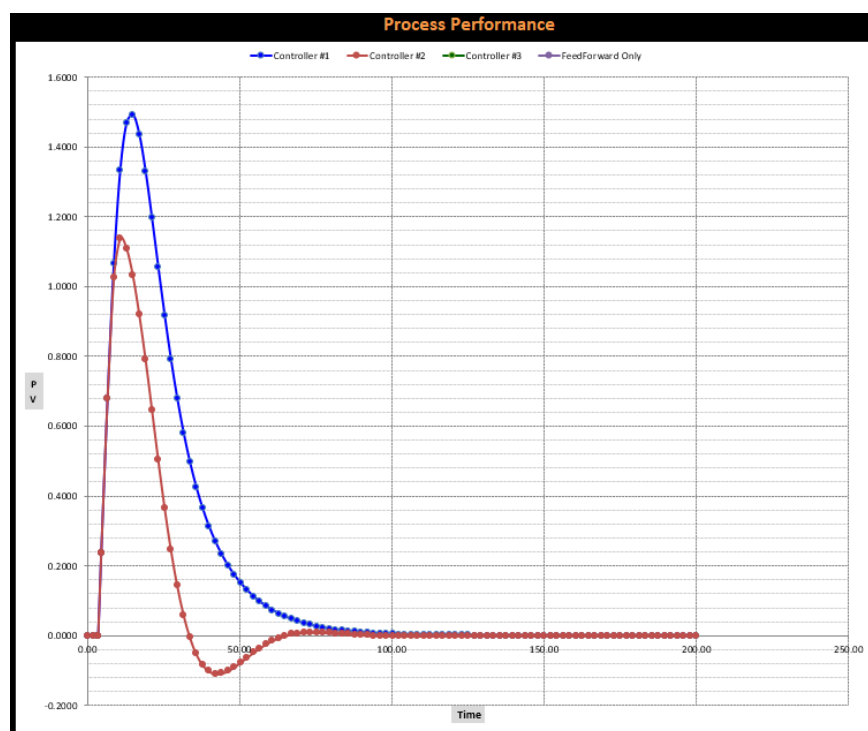


Figure 6: Ziegler-Nichols Method for Feed Temperature Disturbance

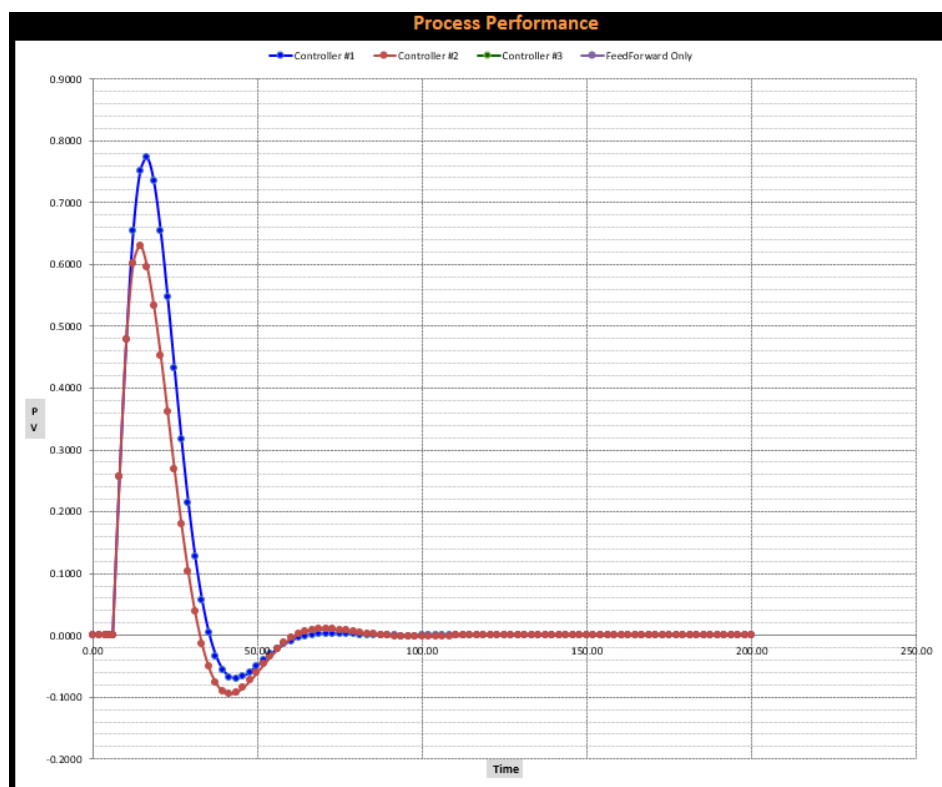


Figure 7: Cohen-Coon Method for Cooling Water Temperature Disturbance

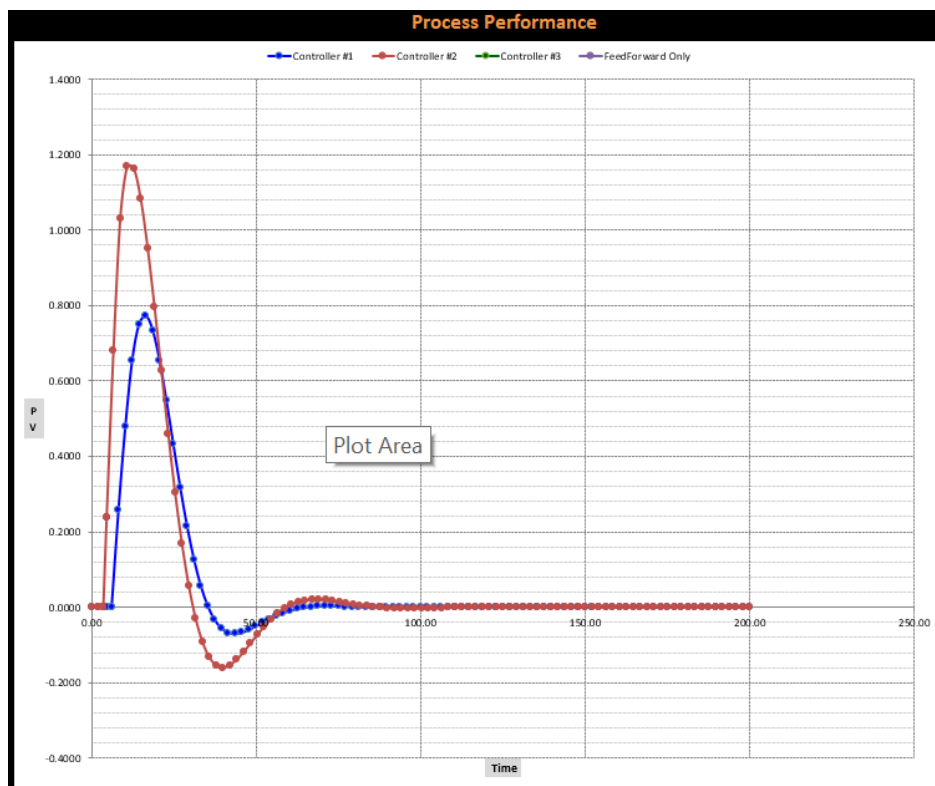


Figure 8: Cohen-Coon Method for Feed Temperature Disturbance

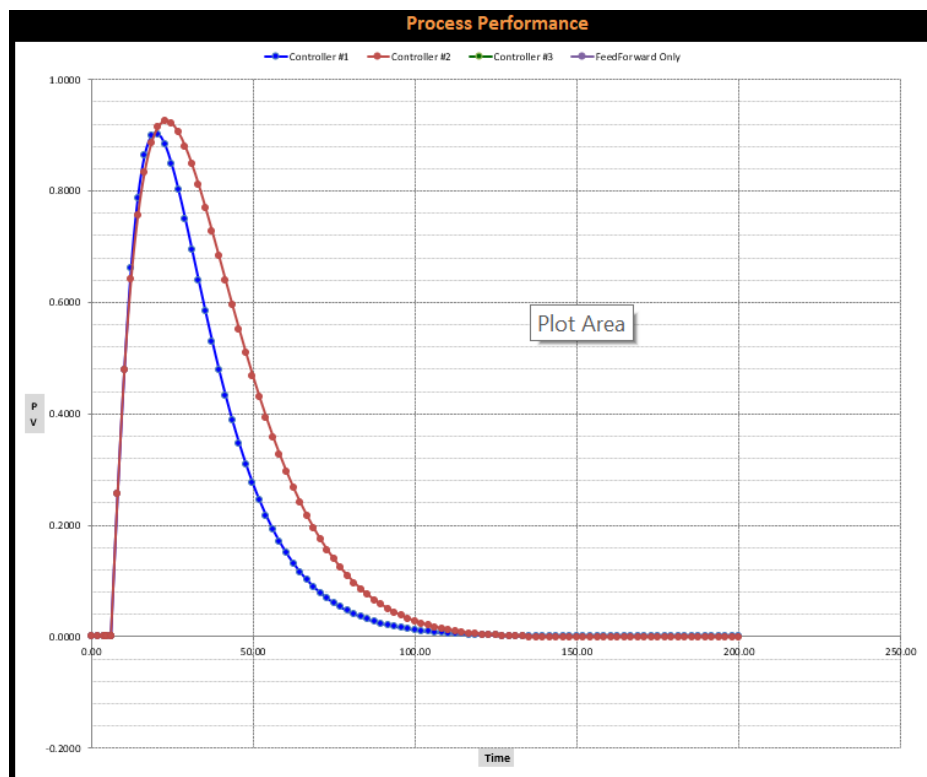


Figure 9: Lambda Tuning Method for Cooling Water Temperature Disturbance

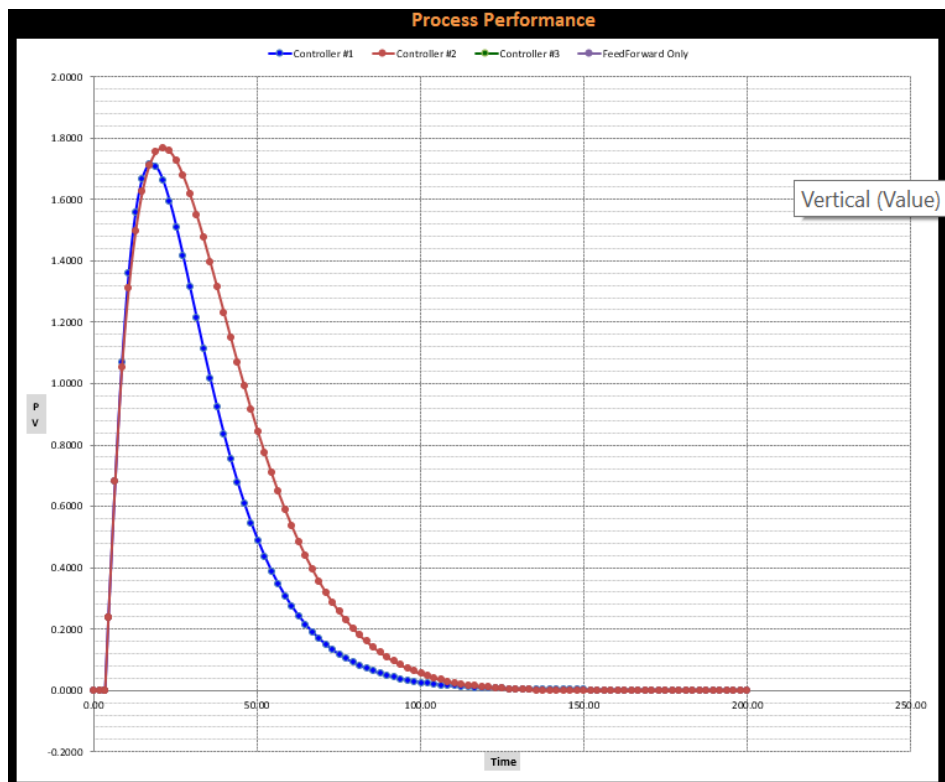


Figure 10: Lambda Tuning Method for Feed Temperature Disturbance

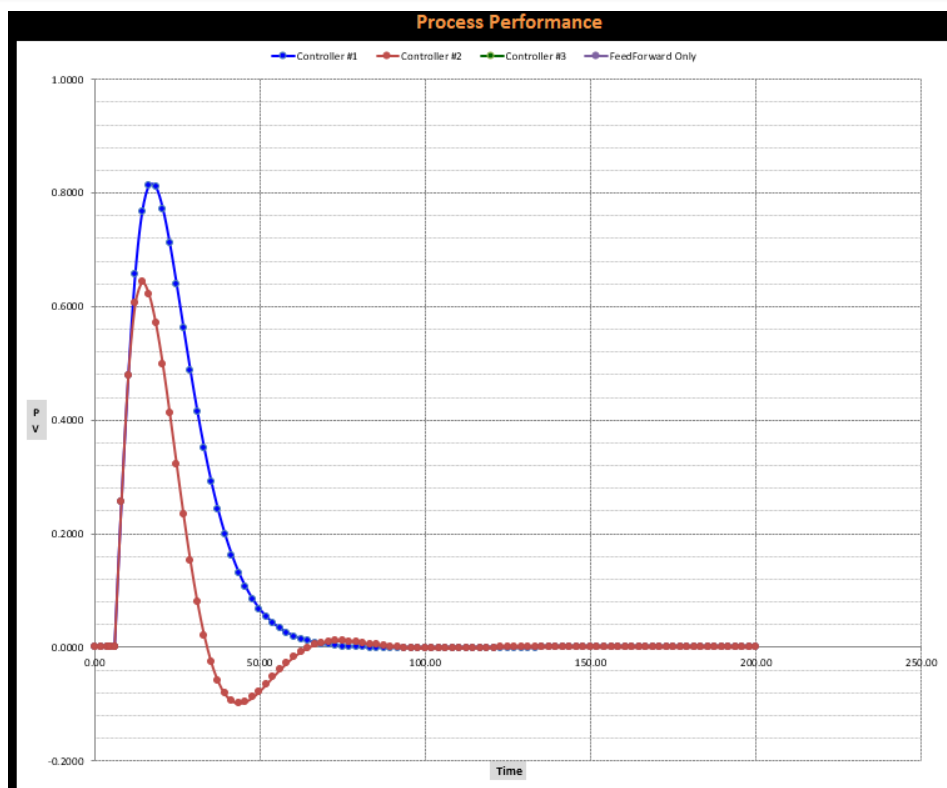


Figure 11: Ultimate Cycling Method for Cooling Water Temperature Disturbance

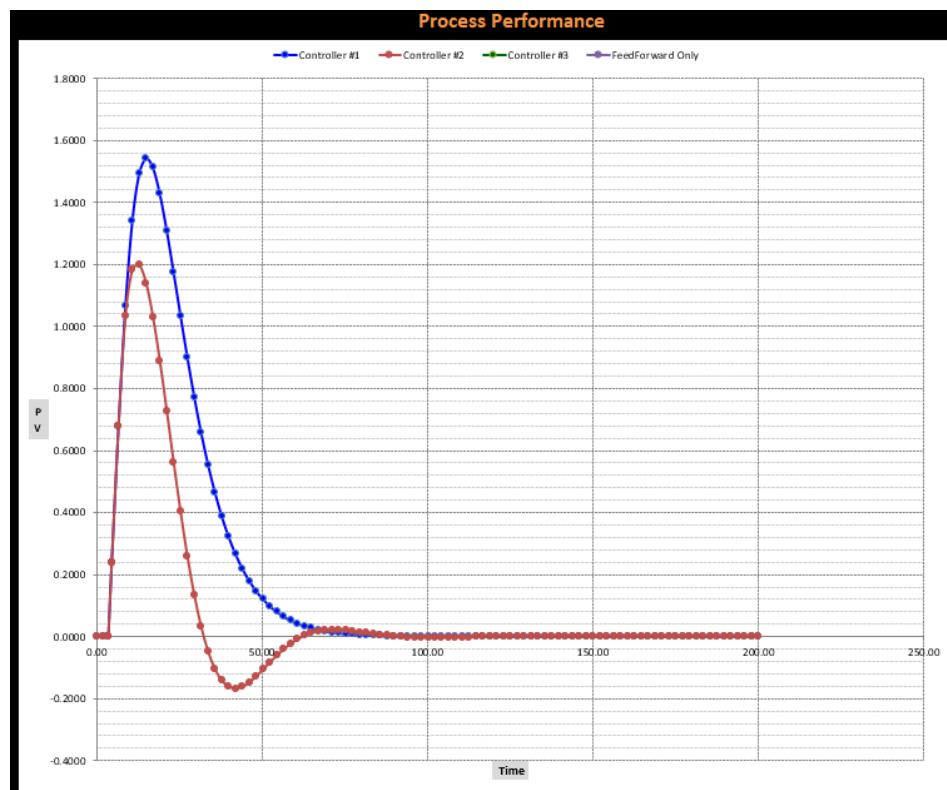


Figure 12: Ultimate Cycling Method for Feed Temperature Disturbance